Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	31 October 2026
Team ID	NA
Project Name	Electric Vehicle Market Performance & Insights
Maximum Marks	4 Marks

Functional Requirements:
Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Use the cleaned EV dataset ev_market_clean.csv (199,999 records, 37 attributes including category, use case, manufacturing country, drivetrain, battery type, price, top speed, acceleration, range, autonomy level, ratings and sales) and connect it to Tableau as an extract.	High
2	Data Processing	Use the cleaned file (0 duplicates, 0 missing values); check data types; create Model Year Label, Autonomy Level and 6 KPI calculated fields; create a Hyper extract.	High
3	Business Analysis	SUM of annual sales by use case and manufacturing country; AVG of price by category, customer rating by battery type, range by model year and drivetrain, top speed and acceleration by category; COUNT of vehicles by drivetrain and autonomy level.	High
4	Business Insights & Analytical Outputs	9 worksheets, a dashboard with a KPI strip and 4 charts, and a 3-point Tableau Story answering the three scenarios.	High
5	User Interaction & Analysis	Filters for Manufacturing Country, Drivetrain and Recommended Use Case applied to all sheets; tooltips; sorted bars; story navigation.	Medium

6	Data Update & Reporting	Dashboard and story published on Tableau Public; refreshing the extract updates every chart.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Title and filters on the left, KPIs on top and two rows of two charts – easy to scan.
NFR-2	**Security**	Only public, non-personal market data; dashboard is read-only on Tableau Public.
NFR-3	**Reliability**	Cleaned data (0 duplicates, 0 missing values); KPI values checked against the source data.
NFR-4	**Performance**	Extract of 199,999 rows; charts update within seconds when filters change.
NFR-5	**Availability**	Available 24×7 through the Tableau Public link.
NFR-6	**Scalability**	New data can be added by refreshing the extract; layout supports more charts.